Homework 9

November 4, 2026

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Problem 1

Let $\mathcal{A} \in L(\mathbb{C}_n[\lambda])$ be defined by $\mathcal{A}f(\lambda) = f(\lambda + a)$, $a \neq 0$. Using its matrix A in the basis $1, \lambda, \ldots, \lambda^n$ from Homework 5, Problem 3, find the minimal polynomial of A.

Solution

$$A_{ij} = \begin{cases} \binom{j}{i} a^{j-i}, & 0 \le i \le j \le n, \\ 0, & i > j, \end{cases} \qquad \Delta = \mathcal{A} - I.$$

$$\Delta(\lambda^k) = (\lambda + a)^k - \lambda^k = ka\lambda^{k-1} + \text{lower-degree terms}, \qquad k \ge 1.$$

$$\Delta^{n+1} = 0, \qquad \Delta^n(\lambda^n) = n!a^n \neq 0 \quad \Longrightarrow \quad (A - I)^{n+1} = 0, \quad (A - I)^n \neq 0.$$

$$\boxed{m_A(z) = (z - 1)^{n+1}}.$$

Problem 2

In a unitary space V, let $\|\alpha\| = \sqrt{\langle \alpha, \alpha \rangle}$. Prove (1) the triangle inequality and (2) the Cauchy–Schwarz inequality (Theorem 4.3).

Solution

Use the course convention: conjugate linearity in the first slot and linearity in the second. Prove (2) first.

$$\beta = 0 \implies |\langle \alpha, \beta \rangle| = 0.$$

For $\beta \neq 0$, set $c = \langle \beta, \alpha \rangle / \|\beta\|^2$. Then

$$\begin{aligned} 0 &\le \|\alpha - c\beta\|^2 \\ &= \|\alpha\|^2 - c\langle \alpha, \beta \rangle - \bar{c}\langle \beta, \alpha \rangle + |c|^2 \|\beta\|^2 \\ &= \|\alpha\|^2 - \frac{|\langle \alpha, \beta \rangle|^2}{\|\beta\|^2}. \end{aligned}$$

$$\boxed{|\langle \alpha, \beta \rangle| \le \|\alpha\| \, \|\beta\|}.$$

(1)

$$\begin{aligned} \|\alpha + \beta\|^2 &= \|\alpha\|^2 + 2\,\mathrm{Re}\langle \alpha, \beta \rangle + \|\beta\|^2 \\ &\le \|\alpha\|^2 + 2|\langle \alpha, \beta \rangle| + \|\beta\|^2 \le (\|\alpha\| + \|\beta\|)^2. \end{aligned}$$

$$\boxed{\|\alpha + \beta\| \le \|\alpha\| + \|\beta\|}.$$

Problem 3

Let $V = \mathbb{R}_n[\lambda]$ and $\langle p, q\rangle = \int_{-1}^{1} p(\lambda)q(\lambda)\,d\lambda$. (1) Prove this is an inner product. (2) Find the Gramian of $1, \lambda, \ldots, \lambda^n$. (3) Apply Gram–Schmidt to obtain an orthonormal basis $\alpha_k L_k(\lambda)$, $0 \le k \le n$.

Solution

(1)

$$\langle p, q\rangle = \langle q, p\rangle, \qquad \langle ap + bq, r\rangle = a\langle p, r\rangle + b\langle q, r\rangle \quad (a, b \in \mathbb{R}),$$

$$\langle p, p\rangle = \int_{-1}^{1} p^2\,d\lambda \ge 0, \qquad \int_{-1}^{1} p^2\,d\lambda = 0 \implies p|_{[-1,1]} = 0 \implies p = 0.$$

(2)

$$\boxed{G = (g_{ij})_{i,j=0}^{n}, \qquad g_{ij} = \int_{-1}^{1} \lambda^{i+j}\,d\lambda = \begin{cases} \dfrac{2}{i+j+1}, & i+j \text{ even}, \\ 0, & i+j \text{ odd}. \end{cases}}$$

(3)

$$q_0 = 1, \qquad q_k = \lambda^k - \sum_{j=0}^{k-1} \frac{\langle q_j, \lambda^k\rangle}{\langle q_j, q_j\rangle} q_j, \qquad e_k = \frac{q_k}{\|q_k\|}.$$

$$q_1 = \lambda, \qquad q_2 = \lambda^2 - \frac{2/3}{2} = \lambda^2 - \frac{1}{3},$$

$$q_3 = \lambda^3 - \frac{2/5}{2/3}\lambda = \lambda^3 - \frac{3}{5}\lambda.$$

Use the standard Legendre normalization $L_k(1) = 1$:

$$L_k(\lambda) = \frac{1}{2^k k!}\frac{d^k}{d\lambda^k}(\lambda^2 - 1)^k, \qquad c_k = \frac{(2k)!}{2^k (k!)^2}.$$

Integrate by parts k times; all endpoint terms vanish.

$$\langle L_k, \lambda^j\rangle = \frac{(-1)^k}{2^k k!}\int_{-1}^{1} (\lambda^2 - 1)^k \frac{d^k \lambda^j}{d\lambda^k}\,d\lambda = 0 \quad (0 \le j < k).$$

$$L_k = c_k \lambda^k + \cdots \quad \Longrightarrow \quad q_k = \frac{L_k}{c_k}.$$

Put $I_k = \int_{-1}^{1} (1 - \lambda^2)^k\,d\lambda$.

$$I_0 = 2, \qquad 0 = \int_{-1}^{1} \frac{d}{d\lambda}\big[\lambda(1 - \lambda^2)^k\big]d\lambda = (2k+1)I_k - 2kI_{k-1} \quad (k \ge 1),$$

$$I_k = \frac{2^{2k+1}(k!)^2}{(2k+1)!}.$$

$$\|L_k\|^2 = \frac{(-1)^k}{2^k k!}\int_{-1}^{1} L_k^{(k)}(\lambda)(\lambda^2 - 1)^k\,d\lambda = \frac{c_k}{2^k} I_k = \frac{2}{2k+1}.$$

$$\boxed{\alpha_k = \sqrt{\frac{2k+1}{2}}, \qquad e_k(\lambda) = \sqrt{\frac{2k+1}{2}}\,L_k(\lambda), \quad k = 0, \ldots, n.}$$

$$e_0 = \frac{1}{\sqrt{2}}, \quad e_1 = \sqrt{\frac{3}{2}}\lambda, \quad e_2 = \sqrt{\frac{5}{8}}(3\lambda^2 - 1), \quad e_3 = \sqrt{\frac{7}{8}}(5\lambda^3 - 3\lambda).$$

Problem 4

Prove Theorems 4.7 and 4.8 in the textbook.

Theorem 4.7. An $n \times n$ matrix A is unitary if and only if its n columns (rows) form an orthonormal basis of $\mathbb{C}^n$.

Solution

$$A = [a_1 \ \cdots \ a_n], \qquad (A^*A)_{ij} = a_i^* a_j = \langle a_i, a_j \rangle.$$

$$A^*A = I \iff a_i^* a_j = \delta_{ij}; \qquad A^*A = I \Longrightarrow A^{-1} = A^* \Longrightarrow AA^* = I.$$

The row statement follows from $AA^* = I$ in the same way; n orthonormal vectors form a basis.

Theorem 4.8. Let $\mathcal{U} \in L(V)$. The following are equivalent:

(1) $\mathcal{U}$ is unitary;

(2) $\|\mathcal{U}\alpha\| = \|\alpha\|$, $\forall \alpha \in V$;

(3) $\mathcal{U}$ maps an orthonormal basis to an orthonormal basis;

(4) the matrix of $\mathcal{U}$ in an orthonormal basis is unitary.

Solution

$$(1) \Longrightarrow (2): \quad \|\mathcal{U}\alpha\|^2 = \langle \mathcal{U}\alpha, \mathcal{U}\alpha \rangle = \langle \alpha, \alpha \rangle.$$

$$\begin{aligned}(2) \Longrightarrow (1): \quad 4\langle \alpha, \beta \rangle &= \|\alpha + \beta\|^2 - \|\alpha - \beta\|^2 \\ &\quad + i\|\alpha - i\beta\|^2 - i\|\alpha + i\beta\|^2 \\ &= 4\langle \mathcal{U}\alpha, \mathcal{U}\beta \rangle.\end{aligned}$$

Over $\mathbb{R}$, use only the first two terms. For an orthonormal basis $E = (e_1, \ldots, e_n)$,

$$(1) \Longrightarrow (3): \quad \langle \mathcal{U}e_i, \mathcal{U}e_j \rangle = \delta_{ij}.$$

$$(3) \Longrightarrow (1): \quad \langle \mathcal{U}\sum_i x_i e_i, \mathcal{U}\sum_j y_j e_j \rangle = \sum_{i,j} \bar{x}_i y_j \delta_{ij} = \sum_i \bar{x}_i y_i = \langle \sum_i x_i e_i, \sum_j y_j e_j \rangle.$$

$$C = [\mathcal{U}]_E: \quad \langle \mathcal{U}\alpha, \mathcal{U}\beta \rangle = x^* C^* C y, \qquad \langle \alpha, \beta \rangle = x^* y \quad \Longrightarrow \quad (1) \iff C^*C = I \iff (4).$$

Problem 5

Let $A \in \mathbb{C}^{n \times n}$ and $r = \operatorname{rank} A$. Prove $A = A^* = A^2$ if and only if $A = UU^*$ for some $U \in \mathcal{U}^{n \times r}$, where $U^*U = I_r$.

Solution

($\Leftarrow$)

$$A^* = (UU^*)^* = UU^* = A, \qquad A^2 = U(U^*U)U^* = UU^* = A, \qquad \operatorname{rank} A = \operatorname{rank} U = r.$$

($\Rightarrow$)

$$S = \operatorname{im} A, \quad T = \ker A, \qquad x = Ax + (I - A)x \in S + T, \qquad S \cap T = \{0\}.$$

$$z \in T: \quad (Ay)^* z = y^* A^* z = y^* A z = 0 \quad \Longrightarrow \quad T = S^{\perp}.$$

Choose an orthonormal basis of S and set $U = [u_1 \ \cdots \ u_r]$.

$$U^*U = I_r, \qquad Ax \in S \Longrightarrow Ax = UU^*Ax, \qquad (I - A)x \in S^{\perp} \Longrightarrow U^*Ax = U^*x.$$

$$Ax = UU^*x \quad (\forall x) \quad \Longrightarrow \quad \boxed{A = UU^*}.$$

For $r = 0$, take the $n \times 0$ empty matrix U; then $UU^* = 0 = A$.